Master Documentation — Spread Reversion & Pairs Trading System

Everything we know so far, consolidated. Strategies, code, data, findings, capital reality, and the path forward.

Date: 2026-08-29 **Scope:** NSE ETF pair trading (Groww) + crypto perpetual pairs trading (Hyperliquid)
Author: Trading research project (user + AI-assisted analysis)

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1. Mission & Goal

Primary objective: **earn money with minimal capital (₹10,000–20,000)**.

Secondary objective: build a systematic, research-backed trading stack (paper → live) for statistical arbitrage / pairs trading.

The core tension (proven repeatedly): mean-reversion strategies return ~10–16%/yr. To earn **₹10,000/month** (₹1.2L/yr) you need ~₹9.6–10.5L of capital. At ₹10–20k, that target requires a 600–1200%/yr return — **not achievable by any legitimate strategy**. The gap is capital, not cleverness.

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2. The Strategy — Statistical Arbitrage (Pole 2007)

Concept: two instruments that normally move together sometimes diverge. Bet the gap snaps back (mean reversion).

- **Spread:** `s = ln(PA) − β·ln(PB)`, with `β = 1.0` (unit-neutral pair: buy 1 A, sell 1 B).
- **Rolling z-score:** `z = (s − mean) / std` over a trailing window.
- **Entry:** `|z| > 2` (gap is unusually wide).
- `z > +2` → **SHORT** the spread (sell A / buy B) — bet the gap narrows.
- `z < −2` → **LONG** the spread (buy A / sell B) — bet the gap widens back.
- **Exit:** z crosses 0 (gap returns to its average). Optional stop-loss / take-profit / confidence gate.

Pairs: - Primary: BANKBEES (Nippon Bank Bees) vs SETFNIFBK (SBI Nifty Bank ETF) — both track Nifty Bank. - Secondary: NIFTYBEES vs SETFNIF50. - Crypto candidates: see §9.

Important: this is *statistical* arbitrage — **probabilistic, not risk-free**. The gap can widen and stay wide. Gross win rate is high (~72–100%) but losses happen, and fees can erase the edge.

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3. Venue Comparison: Groww (NSE) vs Hyperliquid (Crypto)

Factor	Groww / NSE	Hyperliquid (crypto perps)
Fees	■20/order or 0.1% (min ■**5/leg**), + STT/stamp/exchange/SEBI/IPFT/GST	**0.045% taker, 0.015% maker**, no per-order floor
Pair round trip	~■24–133 (4 legs)	~0.09–0.3% of notional
Shorting	Intraday only (MIS), square-off EOD	24/7, symmetric, no borrow
Position size	Whole ETF units (■500–600/unit)	Arbitrary, sub-$10 possible
Universe	~12 relevant index ETFs	232 perps
Capital friendliness	**Poor at small size** (fee floor)	**Good** (tiny fees + leverage)
India tax	STT/stamp built in; capital gains rules	30% VDA, no loss set-off (■■ see §10)

Bottom line: For ■10–20k, **Hyperliquid is the only venue where small capital isn't structurally killed by fees.** The NSE/Groww route needs ■1.2L+ to be net-profitable on pairs.

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4. What We Built — Code Inventory

Python (deployable to Groww, or runnable locally)

File	Purpose	Status
`main.py`	Paper trader — fees modeled, `PAPER_TRADING` toggle, stdout-only (sandbox-safe)	Working
`live_trader.py`	Real-money trader — MARKET/MIS orders, margin-sized from API, sanity checks	Working (needs token)

`portfolio_trader.py`	Paper trader — ■ portfolio sizing + stop-loss/take-profit	Working
`indicator_groww.py`	Live two-sided indicator, 1–5s refresh, **alert sound**, qty suggestion	Working (needs token)
`live_trade.py`	One-sided (LONG) live trader — alerts + **live stop-loss display**	Working (needs token)
`get_access_token.py`	Token generator	**■■ Contains credentials — do not commit/upload**

Backtests (`backtesting/`)

File	Data	What it tests
`backtest.py`	Groww historical	Original Groww-data backtest
`backtest_yahoo.py`	Yahoo 5m (60d)	Intraday, qty sweep
`backtest_portfolio_yahoo.py`	Yahoo daily 2023–26	81-config stress grid
`backtest_report_yahoo.py`	Yahoo daily 2023–26	Full metrics (Sharpe, MDD, …) + MD report
`backtest_one_side.py`	Yahoo (any pair, 5m or daily)	**Parameterized** one-sided LONG + stop-loss
`backtest_master.py`	Yahoo (intraday or daily)	Two-sided, adjustable ■ portfolio, full metrics + MD
`indicator_live.py`	Yahoo 5m	Live-refresh LONG-only manual indicator

Generated outputs: `spread_reversion_yahoo.png`, `portfolio_stress_yahoo.png`, `report_equity.png`, `master_equity.png`, `backtest_one_side_*.png`, `report.md`, `master_report.md`, `report_layman.md`, `indicator_live.png`.

Rust (the original engine + additions)

File	Purpose
`src/main.rs`	Entry point; added `--indicator` mode branch (existing engine untouched)
`src/groww.rs`	Groww API client (token + quote)
`src/trading.rs`	Paper-trading strategy engine (30s poll, window 60)
`src/indicator.rs`	**New** — live two-sided indicator (1–5s, alerts, env-configurable)

`src/bin/simulator.rs`	**New** — market-replay simulator: loads stored CSV, replay from any/random start, **confidence score**, confidence-gated trades, **benchmarks optimal universal position %**
`src/tui.rs`, `src/state.rs`, `src/logging.rs`, `src/notify.rs`, `src/ledger.rs`	Existing engine support

Run indicator: `cargo run -- --indicator` · Simulator: `cargo run --bin simulator -- --data file.csv --benchmark`

Research / studies (external dir)

`/home/null/Documents/studies/trading/` — books, income study guide, ML pairs-trading research, **Hyperliquid screener + backtest** (see §9).

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5. Groww API & Fee Facts

API rules (from AGENTS.md)

- **Only two endpoints allowed:** token generation (`POST /v1/token/api/access`) and live quote (`GET /v1/live-data/quote`). (SDK also exposes `get_ltp`, margin, order APIs.)
- **Paper trading only / NO real orders** unless using `live_trader.py` deliberately.
- Fetch data only for held tokens (BANKBEES, SETFNIFBK).

Rate limits

Type	Per sec	Per min
Authentication (token)	5	30
Orders	10	250
Live Data	**10**	**300**
Non-trading (margin, positions)	20	500

→ A **1–5s refresh** of `get_ltp` (1 call fetches both symbols) = 12–60 calls/min, **well within limits**.

Sandbox constraints (Groww deployment)

- **Read-only filesystem** — no file writes (no log/state files). Stdout-only logging.
- Runs the uploaded script as `/tmp/code/main.py`.

Fee model (intraday MIS, per order)

Component	Rate	Applied
Brokerage	■20 or 0.1% of value, **min ■5**	both sides

STT	0.025%	SELL (intraday)
Stamp duty	0.003%	BUY
Exchange (NSE)	0.00297%	both
SEBI	0.0001%	both
IPFT	0.0001%	both
GST	18% on (brokerage + exchange + SEBI + IPFT)	both

Round-trip cost (4 legs) at ■550 ETF: | qty | Notional/leg | Fees/round trip | |---|---|---| | 2 | ■1,100 | ■**24** | | 19 | ■10,450 | ■**61** | | 100 | ■55,000 | ■**133** |

API gotchas discovered

- `get_order_margin_details` **rejects basket orders for CASH segment** → must call per-leg.
- `place_order` rapid-fire rejection → add ~0.5s between legs.
- `resp.get(key, default)` needed (missing keys), plus price range validation.

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6. Key Findings (Data-Backed)

6.1 The fee floor is the #1 problem

The BANKBEES/SETFNIFBK edge is only **~■1.4/unit** per reversion. Fees are ~■24+ per round trip no matter how small. **At ■5–20k (qty 2–17), fees are 4–6× the edge → guaranteed net loss.**

6.2 "est_gross" is misleading

Live session showed `est_gross = +■103.61` but the account **lost money** — est_gross is LTP-based, pre-fees, pre-slippage. Real result: **net loss ≈ –■90 + fees**. Always report net after fees.

6.3 The pair IS high-confidence (the edge is real)

- Gross win rate ~72–100% on the ETF pair (matches AGENTS.md "73.4%").
- Intraday 60d gross: **+■9,491 at qty 100** — reversion genuinely works.
- **Fees, not reliability, are what fail at small capital.**

6.4 Confidence gating helps

On daily data, one-sided LONG: `conf_min=0` → net –■118; **`conf_min=40`** → **net +■96** (fewer, higher-quality trades). The indicator's confidence score is doing real work.

6.5 Wider pairs fail the reliability test

HDFCBANK/KOTAKBANK (bigger dislocations) lost **intraday**: 74 trades, gross –■**1,053**, 53% win. The daily "+■1,524" was a 7-trade small-sample artifact. **Bigger dislocations ≠ reliable reversion.**

6.6 Stop-loss is inert on daily data

On daily bars the spread reverts before the ■50–100 stop is ever hit (all exits are z-cross). Stop-loss matters intraday / at larger size.

6.7 The Rust simulator validated the confidence concept

Gating on confidence flipped one-sided daily from net-negative to net-positive, and the benchmark sweep (position % 5→95) reports the optimal universal size.

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7. Backtest Results Summary

Intraday (Yahoo 5m, ~60 days, 109 trades, two-sided)

qty	Gross ■	Fees ■	Net ■	Net win%
2 (■5k)	+190	2,664	–**2,474**	0%
100	+9,491	14,863	–5,372	8%
200	+18,982	19,436	–454	40%
300	+28,473	24,010	**+4,463**	57%
500	+47,455	33,156	**+14,299**	72%

→ Net positive needs **qty** ≥ **~250–300** (■3–6L intraday notional).

Daily (2023–2026, 3.6 years, ~36 trades)

Config	Net ■ (■5k)	Final ■
Base (w60,z2,SL100)	–732	4,268
Best (w120,z3)	–151	4,849
Worst (w20,z2)	–1,118	3,882

All 81 grid configs net-negative at ■5k. **At qty=100 daily: net +■3,848** (positive!). Sharpe base –3.04, MaxDD –14.7%.

One-sided LONG (■20k, 5m, 60d) — with stop-loss

40 trades, 0 stops hit, gross +■560, fees ■2,031, net –■1,471. Same fee story.

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8. Financial Reality & Capital Requirements

To earn ■10,000/month (■1.2L/yr): | Source / Strategy | Annual Return | Capital needed | |---|---|---| | Value investing (Graham) | ~12% | ■10L | | Pairs trading (Pole) | ~12.5% | ■9.6L | | Quant trading (Chan) | ~10% | ■12L | | ETF 50-50 | ~12% | ■10L | | **Hyperliquid crypto perps** | ~16% gross (after 30% VDA tax) | ■10.5L |

There is exactly ONE low-capital exception claimed: "stat arb bot @ ■25k MIS margin" — but our backtests prove that claim is based on gross win rate and is net-negative after fees**. Do not fund it expecting ■6–30k/month.**

Honest summary: - ■10–20k → ■10k/month = 600–1200%/yr → **impossible safely** - ■10–20k → **learning + modest growth + building capital** = realistic - To make pairs trading actually pay, either raise capital to ■1.2L+ (NSE) or use Hyperliquid (small-capital-friendly venue)

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9. Hyperliquid Pairs Research (Crypto)

This is the **most promising small-capital path**. Full stack exists in the studies dir.

Why it fits

- Fees: taker 0.045%, maker 0.015% (no ■5 floor). Round trip ~0.1–0.3%.
- Perps: short any time, arbitrary size, 24/7, no borrow.
- Small accounts are genuinely advantaged where institutions can't trade (funding dispersion lives on illiquid names).
- API readable from India, no account needed (read access).

Screener (R1–R5 filter, 232 perps)

1. **R1** Cointegration (Engle-Granger, p<0.01, both directions)
2. **R2** Hurst exponent < 0.5 (mean-reverting)
3. **R3** Half-life 1–60 days
4. **R4** ≥6 mean-crossings/year
5. **R5** Live bid-ask must clear round-trip costs by ≥3×

Funnel: 4,005 candidate pairs → 173 (R1) → 47 (R2–R4). **47 pairs cleared all filters** ($250k/24h vol, ≥365-day maturity).

Walk-forward backtest (40 testable pairs, 26 profitable)

| Pair | Return | Sharpe | MaxDD | Notes | |---|---|---:|---:|---:|---| | **BSV/FIL** | **+391%** | 2.82 | –7.4% | 7 trades, 0 stops — cleanest | | MINA/FIL | +231% | 1.17 | –28.9% | | | KAS/PEOPLE | +180% | 1.12 | –43.8% | | | BSV/ICP | +161% | 1.48 | –27.1% | | | FIL/ICP | +133% | 0.91 | –36.7% | | | RUNE/POPCAT | +79% | 1.43 | –21.7% | | | BSV/S | +20% | 2.08 | –9.6% | |

Portfolio (equal-weight all testable pairs): total **+202%**, Sharpe **1.0**, MaxDD –**38.6%**.

Honest warnings: - **64% of all 476 simulated trades exited via stop-loss**, not clean reversion. - Many pairs have –50 to –77% MaxDD (LTC/XRP, kSHIB/FIL, ADA/PYTH, SAND/IOTA) — **avoid these**. - Pairs

are selected on the same history they're tested on (flatters results). - VIRTUAL/PAXG (best on paper) **never even triggered a trade** — "looks great" ≠ "makes money".

Tradeable shortlist for small capital (suggested): BSV/FIL, MINA/FIL, BSV/ICP, RUNE/POPCAT, BSV/S, kSHIB/SEI — high Sharpe, low MDD, few stop-outs.

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10. Tax & Legal (India)

- **Crypto (VDA):** 30% flat tax on gains, **no loss set-off**, 1% TDS possible per trade — this alone roughly halves crypto edge (guide: ~16% gross → ~9% effective).
- **Hyperliquid perps classification** (VDA vs business income) and **offshore DEX custody risk** are open questions — get a CA's written answer **before funding** (this is the kill criterion in the research plan).
- **NSE/Groww:** STT/stamp/exchange fees are auto-deducted; standard capital-gains rules apply.

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11. Risk Management Rules

1. **Never risk more than 1% of total equity on a single trade** (Hite/Kovner rule).
2. Position size from account equity, not from "gut".
3. Use stop-losses; expect ~64% of trades to hit them in crypto pairs — size for that.
4. Only take high-confidence signals (confidence gate ≥ 40 improves results).
5. Paper-trade first; **fund only if net Sharpe ≥ 1.0 after 3 months**.
6. Cap leverage — the basket backtest shows –38.6% MDD even unlevered.
7. Keep fees front-of-mind: at small size, every ■ of fee matters.
8. Treat "■10k/month from ■10k" as a red flag — that's a scam-level return claim.

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12. Recommendations & Next Steps

Verdict

- **NSE/Groww pair bot at ■10–20k: dead end** (fee floor). Only viable at ■1.2L+ (intraday) or ■1.2L (daily qty 100).
- **Hyperliquid pairs: the genuine small-capital venue** — but it's high-variance and tax-eroded. Treat as *learn + grow + build capital*, not ■10k/month.

Phased plan

Phase	Action	Kill criterion
0	Fix the income-guide's false "■25k → ■6–30k/mo bot" claim (gross-win fantasy)	—
1	**Tax/legal** with a CA (VDA/TDS/custody)	1% TDS per trade → stop

2	Build the small-account **tradeable shortlist** from backtest_results.csv	—
3	**Paper-trade** with honest cost model (fees + funding + 30% tax)	Net Sharpe < 1.0 after 3 mo → don't fund
4	**Minimum live capital** ($100–300) — measure modelled-vs-real fills	Fill gap too large → stop
5	Optionally test **intraday (5m/1h) pairs** for higher frequency (higher risk/return)	—

Realistic expectations

- With ■10–20k: aim for **3–10%/mo high-frequency (high variance)** or **single-digit %/yr conservative**.
- Reaching **■10k/month requires ~■10L** of capital, or ~10× the risk.
- The single highest-leverage change for small capital is the **venue** (Hyperliquid), not the strategy.

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13. File Map / Where Everything Lives

Repo root: `/home/null/Documents/development/statistical_arbitrage/spread_reversion/` - `main.py`, `live_trader.py`, `portfolio_trader.py`, `indicator_groww.py`, `live_trade.py` — Python traders/indicators - `backtesting/` — all backtests + reports + charts - `src/` — Rust engine + `indicator.rs` + `bin/simulator.rs` - `documentation/` — this master doc + PDF - `AGENTS.md`, `.env.example`, `README.md` — config/docs - ■■ `get_access_token.py`, `.env` — **credentials, do not commit/upload**

Studies dir: `/home/null/Documents/studies/trading/` - `Machine_Learning_Base_Pairs_Trading/` — Hyperliquid screener/backtest + reports - `hyperliquid_pairs/` — `pairs_screener.py`, `backtest.py`, `hl_data.py`, `results_final.csv`, `backtest_results.csv`, `backtest_portfolio_equity.csv` - `The_Indian_Market_10000_INR_Monthly_Income_Study_Guide.md` (+ PDF) — master income study - Books (Graham, Chan, Pole, Schwager, Sarmento/Horta) + summaries

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14. Appendix — Fees & Formulas

Gross P&L (pair, unit-neutral)

- **LONG** (buy A, sell B): `gross = (PA_exit − PA_entry)·qty − (PB_exit − PB_entry)·qty`
- **SHORT** (sell A, buy B): `gross = (PA_entry − PA_exit)·qty + (PB_exit − PB_entry)·qty`

z-score

```
spread = ln(PA) − ln(PB) mean = avg(spread window) std = sqrt(var(spread window)) (floor 1e-4 to avoid div-by-zero) z = (spread − mean) / std
```

Confidence score (simulator/indicator)

`confidence = clamp((|z| − 2) / 3 × 100, 0, 100)` — 0 at the entry threshold, 100 at |z|=5.

Groww intraday MIS leg fee

```
fee = min(20, 0.001·V) (min 5) + STT(sell 0.025%V) + stamp(buy 0.003%V) + 0.00297%V +
0.0001%V + 0.0001%V + 18%·(brokerage+exchange+SEBI+IPFT)
```

Hyperliquid fee

`taker 0.045% / maker 0.015% per side` + bid-ask crossing + funding (~0.03%/day flat estimate in backtests).

End of master doc. Maintain as findings evolve.